



BBA APEX



Python List – Complete Documentation with Examples

1 Introduction to List

A **List** in Python is:

- Ordered collection
- Mutable (can be modified)
- Allows duplicate values
- Can store different data types

Syntax:

```
list_name = [element1, element2, element3]
```

```
In [1]: a=[1,2,3,45,5]  
type(a)
```

```
Out[1]: list
```

```
In [28]: aru = [10,11,12, 20, 30, 40]
```

```
Out[28]: []
```

2 Characteristics of List

Ordered

Mutable

Indexed

Allows duplicates

Heterogeneous (multiple data types allowed)

```
In [5]: data = [10, "Python", 3.14, True]  
print(data)
```

```
[10, 'Python', 3.14, True]
```

3 Accessing List Elements

◆ Using Index

```
In [6]: fruits = ["Apple", "Banana", "Mango"]
print(fruits[0])    # First element
print(fruits[-1])   # Last element
```

Apple
Mango

3 List Slicing

```
In [7]: numbers = [1, 2, 3, 4, 5, 6]
# print(numbers[1:4])
print(numbers[:2])
```

[2, 3, 4]
[1, 3, 5]

4 Important List Methods

```
In [61]: nums = [1, 2, 3, 7, 1, 0, 5, 8]
nums.count(1)
```

Out[61]: 2

```
In [60]: nums
```

Out[60]: [1, 2]

```
In [52]: nums.append(4)
print(nums)
```

[1, 2, 3, 7, 1, 0, 5, 4]

```
In [59]: nums.insert(1, 10)
print(nums)
```

[1, 10, 2, 3, 4]

```
In [60]: nums.extend([5, 6])
print(nums)
```

[1, 10, 2, 3, 4, 5, 6]

```
In [47]: nums.remove(10)
print(nums)
```

[1, 2, 3, 4, 5, 6]

```
In [40]: nums.pop()
print(nums)
```

[1, 2, 3, 4, 5]

```
In [74]: numss=[2,4,5,66,7,8]
print(numss.index(5))
```

2

```
In [75]: print(numss.count(2))
```

```
1
```

```
In [ ]: nums.sort()
        print(nums)
```

```
In [76]: nums.reverse()
        print(nums)
```

```
[6, 5, 4, 3, 2, 10, 1]
```

5 Looping Through List

```
In [63]: fruits = ["Apple", "Banana", "Mango"]
        for niyati in fruits:
            print(niyati*3)
```

```
AppleAppleApple
BananaBananaBanana
MangoMangoMango
```

6 List Comprehension

```
In [2]: # squares = [x*x for x in range(1, 6)]
        # print(squares)
        a=[2,4,5,8]
        b=[3,4,2,4]
        a*3
```

```
Out[2]: [2, 4, 5, 8, 2, 4, 5, 8]
```

7 Nested Lists

```
In [18]: a=float(input("enter the marks of python"))
        b=float(input("enter the marks of problem_solving"))
        c=float(input("enter the marks of eco"))
        d=float(input("enter the marks of MIS"))
        total=a+b+c+d
        percentage=total/400*100
        print(percentage)
```

```
52.5
```

8 Built-in Functions with List

```
In [80]: numbers = [10, 20, 30, 40]
        print(len(numbers))
        print(max(numbers))
        print(min(numbers))
        print(sum(numbers))
```

4
40
10
100

Practical Example Program

```
In [81]: marks = []

for i in range(5):
    value = float(input("Enter mark: "))
    marks.append(value)

total = sum(marks)
average = total / len(marks)

print("Marks List:", marks)
print("Total:", total)
print("Average:", average)
```

Marks List: [300.0, 310.0, 324.0, 234.0, 231.0]
Total: 1399.0
Average: 279.8

In []: